

# BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2026-2027



SCHOOL OF COMPUTER APPLICATIONS

## ASSIGNMENT OF

NoSQL and DBaaS 101  
(BCADSN23202)

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Roll No. 1250258490

**Submitted To:**

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# MONGODB STUDENT DATABASE PROJECT

## Complete MongoDB Shell (mongosh) Operations

Database: studentProjectDB | Collection: studentRecords

### PROJECT INFORMATION

Database Name: studentProjectDB

Collection Name: studentRecords

Total initial records: 51 (Roll No. 101-151)

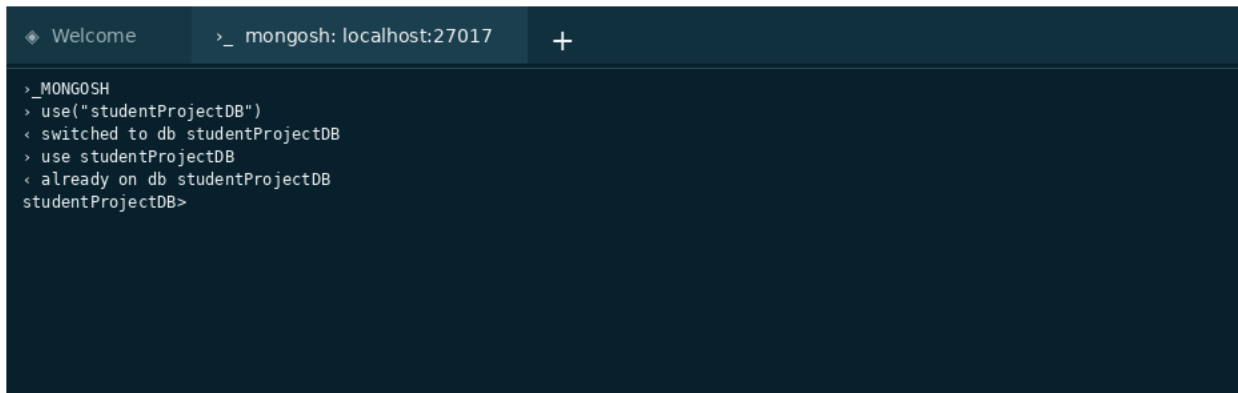
Operations covered: Insert One, Insert Many, Update One, Update Many, Delete One, Delete Many, Query Operations and Verification.

### 1. CREATE / SELECT DATABASE

**Question:** How do you create or select the database for the MongoDB student project?

```
use studentProjectDB;
```

### SCREENSHOT / OUTPUT

A screenshot of the MongoDB Shell (mongosh) interface. The top bar shows a 'Welcome' tab and the current connection '› \_ mongosh: localhost:27017'. The main area displays the following commands and their outputs:

```
> _MONGOSH
> use("studentProjectDB")
< switched to db studentProjectDB
> use studentProjectDB
< already on db studentProjectDB
studentProjectDB>
```

### 2. INSERT ONE

**Question:** Write a MongoDB command to insert one student record into the studentRecords collection.

```
db.studentRecords.insertOne({
  rollNo: 101,
  name: "Aarush",
  age: 18,
  department: "CSE",
  marks: 88
});
```

## SCREENSHOT / OUTPUT

```
> db.studentRecords.insertOne({
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 88
});
{
  acknowledged: true
  insertedId: ObjectId('49da80e19f16f61caba2b07d')
}
studentProjectDB>
```

### 3. INSERT MANY

Question: Write a MongoDB command to insert multiple student records (Roll No. 102-151) into the studentRecords collection.

```
db.studentRecords.insertMany([
  { rollNo: 102, name: "Bhavya", age: 21, department: "IT", marks: 74 },
  { rollNo: 103, name: "Chirag", age: 18, department: "ECE", marks: 85 },
  { rollNo: 104, name: "Diya", age: 21, department: "ME", marks: 63 },
  { rollNo: 105, name: "Eshan", age: 18, department: "CSE", marks: 92 },
  { rollNo: 106, name: "Falguni", age: 21, department: "IT", marks: 79 },
  { rollNo: 107, name: "Gaurav", age: 18, department: "ECE", marks: 36 },
  { rollNo: 108, name: "Hina", age: 21, department: "ME", marks: 95 },
  { rollNo: 109, name: "Ivaan", age: 18, department: "CSE", marks: 68 },
  { rollNo: 110, name: "Jhanvi", age: 21, department: "IT", marks: 81 },
  { rollNo: 111, name: "Keshav", age: 18, department: "ECE", marks: 47 },
  { rollNo: 112, name: "Lavanya", age: 21, department: "ME", marks: 90 },
  { rollNo: 113, name: "Mohit", age: 18, department: "CSE", marks: 72 },
  { rollNo: 114, name: "Nisha", age: 21, department: "IT", marks: 34 },
  { rollNo: 115, name: "Omkar", age: 18, department: "ECE", marks: 86 },
  { rollNo: 116, name: "Prisha", age: 21, department: "ME", marks: 77 },
  { rollNo: 117, name: "Qadir", age: 18, department: "CSE", marks: 58 },
  { rollNo: 118, name: "Ritika", age: 21, department: "IT", marks: 99 },
  { rollNo: 119, name: "Sahil", age: 18, department: "ECE", marks: 65 },
  { rollNo: 120, name: "Trisha", age: 21, department: "ME", marks: 83 },
  { rollNo: 121, name: "Utkarsh", age: 18, department: "CSE", marks: 39 },
  { rollNo: 122, name: "Varun", age: 21, department: "IT", marks: 71 },
  { rollNo: 123, name: "Wamika", age: 18, department: "ECE", marks: 93 },
```

```
{ rollNo: 124, name: "Yash", age: 21, department: "ME", marks: 76 },
{ rollNo: 125, name: "Zoya", age: 18, department: "CSE", marks: 28 },
{ rollNo: 126, name: "Advik", age: 21, department: "IT", marks: 89 },
{ rollNo: 127, name: "Bhumika", age: 18, department: "ECE", marks: 52 },
{ rollNo: 128, name: "Chetan", age: 21, department: "ME", marks: 84 },
{ rollNo: 129, name: "Devika", age: 18, department: "CSE", marks: 67 },
{ rollNo: 130, name: "Farhan", age: 21, department: "IT", marks: 91 },
{ rollNo: 131, name: "Gauri", age: 18, department: "ECE", marks: 73 },
{ rollNo: 132, name: "Harsh", age: 21, department: "ME", marks: 32 },
{ rollNo: 133, name: "Ishani", age: 18, department: "CSE", marks: 87 },
{ rollNo: 134, name: "Jatin", age: 21, department: "IT", marks: 61 },
{ rollNo: 135, name: "Kavya", age: 18, department: "ECE", marks: 78 },
{ rollNo: 136, name: "Lakshay", age: 21, department: "ME", marks: 96 },
{ rollNo: 137, name: "Mira", age: 18, department: "CSE", marks: 69 },
{ rollNo: 138, name: "Nakul", age: 21, department: "IT", marks: 82 },
{ rollNo: 139, name: "Ojas", age: 18, department: "ECE", marks: 44 },
{ rollNo: 140, name: "Pallavi", age: 21, department: "ME", marks: 85 },
{ rollNo: 141, name: "Rohan", age: 18, department: "CSE", marks: 97 },
{ rollNo: 142, name: "Simran", age: 21, department: "IT", marks: 55 },
{ rollNo: 143, name: "Tanish", age: 18, department: "ECE", marks: 80 },
{ rollNo: 144, name: "Uday", age: 21, department: "ME", marks: 70 },
{ rollNo: 145, name: "Vansh", age: 18, department: "CSE", marks: 94 },
{ rollNo: 146, name: "Yuvika", age: 21, department: "IT", marks: 38 },
{ rollNo: 147, name: "Zubin", age: 18, department: "ECE", marks: 66 },
{ rollNo: 148, name: "Aarohi", age: 21, department: "ME", marks: 88 },
{ rollNo: 149, name: "Bhavin", age: 18, department: "CSE", marks: 59 },
{ rollNo: 150, name: "Charu", age: 21, department: "IT", marks: 75 },
{ rollNo: 151, name: "Danish", age: 18, department: "ECE", marks: 84 },
});
```

**SCREENSHOT / OUTPUT**

```

db.studentRecords.insertMany([
  { rollNo: 102, name: 'Bhavya', age: 21, department: 'IT', marks: 74 },
  { rollNo: 103, name: 'Chirag', age: 18, department: 'ECE', marks: 85 },
  { rollNo: 104, name: 'Diya', age: 21, department: 'ME', marks: 63 },
  { rollNo: 105, name: 'Eshan', age: 18, department: 'CSE', marks: 92 },
  { rollNo: 106, name: 'Falguni', age: 21, department: 'IT', marks: 79 },
  { rollNo: 107, name: 'Gaurav', age: 18, department: 'ECE', marks: 36 },
  { rollNo: 108, name: 'Hina', age: 21, department: 'ME', marks: 95 },
  { rollNo: 109, name: 'Ivaan', age: 18, department: 'CSE', marks: 68 },
  { rollNo: 110, name: 'Jhanvi', age: 21, department: 'IT', marks: 81 },
  { rollNo: 111, name: 'Keshav', age: 18, department: 'ECE', marks: 47 },
  { rollNo: 112, name: 'Lavanya', age: 21, department: 'ME', marks: 90 },
  { rollNo: 113, name: 'Mohit', age: 18, department: 'CSE', marks: 72 },
  { rollNo: 114, name: 'Nisha', age: 21, department: 'IT', marks: 34 },
  { rollNo: 115, name: 'Omkar', age: 18, department: 'ECE', marks: 86 },
  { rollNo: 116, name: 'Prisha', age: 21, department: 'ME', marks: 77 },
  { rollNo: 117, name: 'Qadir', age: 18, department: 'CSE', marks: 58 },
  { rollNo: 118, name: 'Ritika', age: 21, department: 'IT', marks: 99 },
  { rollNo: 119, name: 'Sahil', age: 18, department: 'ECE', marks: 65 },
  { rollNo: 120, name: 'Trisha', age: 21, department: 'ME', marks: 83 },
  { rollNo: 121, name: 'Utharsh', age: 18, department: 'CSE', marks: 39 },
  { rollNo: 122, name: 'Vansh', age: 21, department: 'IT', marks: 71 },
  { rollNo: 123, name: 'Wamika', age: 18, department: 'ECE', marks: 93 },
  { rollNo: 124, name: 'Yash', age: 21, department: 'ME', marks: 76 },
  { rollNo: 125, name: 'Zoya', age: 18, department: 'CSE', marks: 28 },
  { rollNo: 126, name: 'Advik', age: 21, department: 'IT', marks: 89 },
  { rollNo: 127, name: 'Bhumika', age: 18, department: 'ECE', marks: 52 },
  { rollNo: 128, name: 'Chetan', age: 21, department: 'ME', marks: 84 },
  { rollNo: 129, name: 'Devika', age: 18, department: 'CSE', marks: 67 },
  { rollNo: 130, name: 'Farhan', age: 21, department: 'IT', marks: 91 },
  { rollNo: 131, name: 'Gauri', age: 18, department: 'ECE', marks: 73 },
  { rollNo: 132, name: 'Harsh', age: 21, department: 'ME', marks: 32 },
  { rollNo: 133, name: 'Ishani', age: 18, department: 'CSE', marks: 87 },
  { rollNo: 134, name: 'Jatin', age: 21, department: 'IT', marks: 61 },
  { rollNo: 135, name: 'Kavya', age: 18, department: 'ECE', marks: 78 },
  { rollNo: 136, name: 'Lakshay', age: 21, department: 'ME', marks: 96 },
  { rollNo: 137, name: 'Mira', age: 18, department: 'CSE', marks: 69 },
  { rollNo: 138, name: 'Nakul', age: 21, department: 'IT', marks: 82 },
  { rollNo: 139, name: 'Ojas', age: 18, department: 'ECE', marks: 44 },
  { rollNo: 140, name: 'Pallavi', age: 21, department: 'ME', marks: 85 },
  { rollNo: 141, name: 'Rohan', age: 18, department: 'CSE', marks: 97 },
  { rollNo: 142, name: 'Simran', age: 21, department: 'IT', marks: 55 },
  { rollNo: 143, name: 'Tanish', age: 18, department: 'ECE', marks: 80 },
  { rollNo: 144, name: 'Uday', age: 21, department: 'ME', marks: 70 },
  { rollNo: 145, name: 'Vansh', age: 18, department: 'CSE', marks: 94 },
  { rollNo: 146, name: 'Yuvika', age: 21, department: 'IT', marks: 38 },
  { rollNo: 147, name: 'Zubin', age: 18, department: 'ECE', marks: 66 },
  { rollNo: 148, name: 'Aarohi', age: 21, department: 'ME', marks: 88 },
  { rollNo: 149, name: 'Bhavini', age: 18, department: 'CSE', marks: 59 },
  { rollNo: 150, name: 'Charu', age: 21, department: 'IT', marks: 75 },
  { rollNo: 151, name: 'Danish', age: 18, department: 'ECE', marks: 84 },
]);
{
  acknowledged: true
  insertedIds: {
    '0': ObjectId('f8ebaa9990f45b8f0420198f'),
    '1': ObjectId('6722edfdd63466f57afb8894'),
    '2': ObjectId('fca42f158438cf44fb2ae930ec'),
    '3': ObjectId('a2b594e246e7613a59509acc'),
    '4': ObjectId('c5ec59fbd922a21fe035b661'),
    '5': ObjectId('a4c56bd6a64df3d1da92f48f'),
    '6': ObjectId('8f41789677c20ccb63df82a1'),
    '7': ObjectId('9890f5644ee53226be7b274'),
    '8': ObjectId('9d7b687bc34afa032fbaed18'),
    '9': ObjectId('5955602abdcac3807bbc0a88'),
    '10': ObjectId('e36bd19c96084e4bd3d33a73'),
    '11': ObjectId('ba2de915d0ba162e9fe8cf2c'),
    '12': ObjectId('0fce807a9808aa06ad2a78c9'),
    '13': ObjectId('5ce305027a3134fb22587ff7'),
    '14': ObjectId('2715c20794c0e0b8e47ce83'),
    '15': ObjectId('f1ca2d80efa580ef6ad7f940'),
    '16': ObjectId('df8a188247a735f778cd5a1a'),
    '17': ObjectId('43c2436ef1697af5a4ec015e'),
    '18': ObjectId('3c585128379498327137be78'),
    '19': ObjectId('143aab4d7eedd6ddfd50e306'),
    '20': ObjectId('d63cdf85782489fc577855e'),
    '21': ObjectId('1ab105bd7793b892140d8592'),
    '22': ObjectId('9b90e242ab0b01bfa45e9d10'),
    '23': ObjectId('51b4d82c82890a6ec04d6cd1'),
    '24': ObjectId('94a04e74e58a68a56be1cd'),
    '25': ObjectId('d7c6a3bd0d9ef687053a36c9'),
    '26': ObjectId('2a01978b73de2ab6a0bc400e'),
    '27': ObjectId('be82c5a6597c13b3826512b'),
    '28': ObjectId('ed8f9f153d361e60711e0dc'),
    '29': ObjectId('5c7e92897545a389a970241c'),
    '30': ObjectId('97dde30b2f626a319ba093ac'),
    '31': ObjectId('b5a1f4b187445ce07d41a2e0'),
    '32': ObjectId('a94d47d9c93fc578fc818858'),
    '33': ObjectId('79a66d8cde6964e812e6216d'),
    '34': ObjectId('057c1c5fd606694a52f6bb97'),
    '35': ObjectId('26aa175f5e031c5371664313'),
    '36': ObjectId('2827abfc08aa590d60487a6f'),
    '37': ObjectId('2cb506ad8457c00f3630ab46'),
    '38': ObjectId('34374b00aee2791a5d67db37')
  }
}

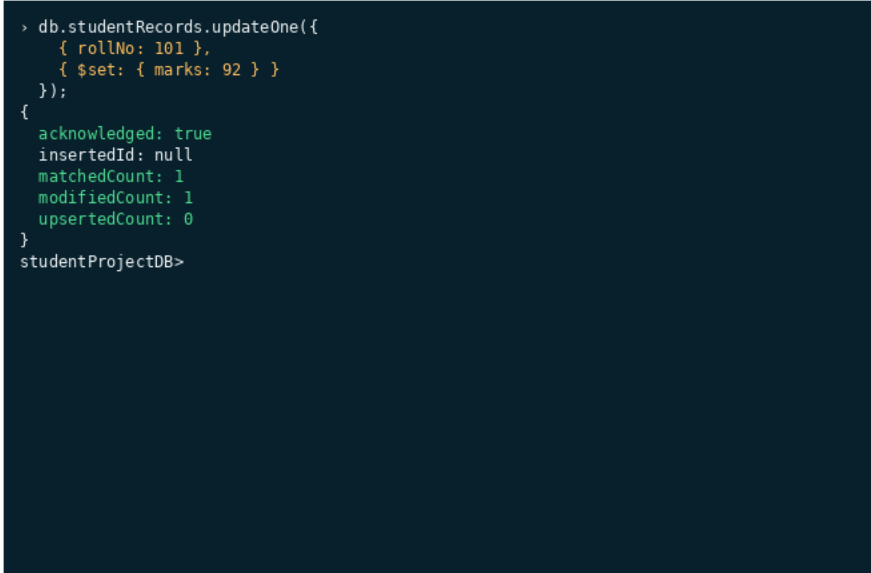
```

## 4. UPDATE ONE

Question: How do you update the marks of the student with Roll No. 101 from 88 to 92?

```
db.studentRecords.updateOne(  
  { rollNo: 101 },  
  { $set: { marks: 92 } }  
);
```

### SCREENSHOT / OUTPUT



```
> db.studentRecords.updateOne(  
  { rollNo: 101 },  
  { $set: { marks: 92 } }  
));  
{  
  acknowledged: true  
  insertedId: null  
  matchedCount: 1  
  modifiedCount: 1  
  upsertedCount: 0  
}  
studentProjectDB>
```

## 5. UPDATE MANY

Question: How do you add the status "Active" to all students belonging to the CSE department?

```
db.studentRecords.updateMany(  
  { department: "CSE" },  
  { $set: { status: "Active" } }  
);
```

### SCREENSHOT / OUTPUT

```
> db.studentRecords.updateMany({ department: "CSE" },
  { $set: { status: "Active" } }
);
{
  acknowledged: true
  insertedId: null
  matchedCount: 13
  modifiedCount: 13
  upsertedCount: 0
}
studentProjectDB>
```

## 6. DELETE ONE

Question: How do you delete the student record having Roll No. 151?

```
db.studentRecords.deleteOne({
  rollNo: 151
});
```

### SCREENSHOT / OUTPUT

```
> db.studentRecords.deleteOne({
  rollNo: 151
});
{
  acknowledged: true
  deletedCount: 1
}
studentProjectDB>
```

## 7. DELETE MANY

Question: How do you delete all student records having marks less than 40?

```
db.studentRecords.deleteMany({
  marks: { $lt: 40 }
});
```

```
> db.studentRecords.deleteMany({ marks: { $lt: 40 } });
{
  acknowledged: true
  deletedCount: 6
}
studentProjectDB>
```

## 8. QUERY OPERATIONS

### A. MARKS GREATER THAN 80

Question: How do you find all students whose marks are greater than 80? (\$gt)

```
db.studentRecords.find({
  marks: { $gt: 80 }
});
```

SCREENSHOT / OUTPUT

```
> db.studentRecords.find({ marks: { $gt: 80 } });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('8f41789677c20cbb63df82a1'),
  rollNo: 108,
  name: 'Hina',
  age: 21,
  department: 'ME',
  marks: 95
}
{
  _id: ObjectId('9d7b687bc34afa032fbead18'),
  rollNo: 110,
  name: 'Jhanvi',
  age: 21,
  department: 'IT',
  marks: 81
}
Type 'it' for more
studentProjectDB>
```

## B. MARKS LESS THAN 80

Question: How do you find all students whose marks are less than 80? (\$lt)

```
db.studentRecords.find({
  marks: { $lt: 80 }
});
```

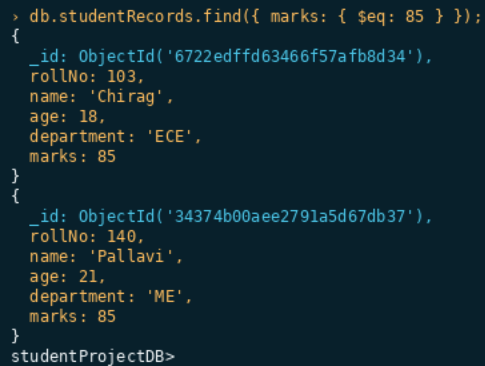
```
> db.studentRecords.find({ marks: { $lt: 80 } });
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
{
  _id: ObjectId('c5ec59fbd922e21fe035b661'),
  rollNo: 106,
  name: 'Falguni',
  age: 21,
  department: 'IT',
  marks: 79
}
{
  _id: ObjectId('9890f5644ee5c3226be7b274'),
  rollNo: 109,
  name: 'Ivaan',
  age: 18,
  department: 'CSE',
  marks: 68,
  status: 'Active'
}
{
  _id: ObjectId('5955602abdcac3807bbc0a88'),
  rollNo: 111,
  name: 'Keshav',
  age: 18,
  department: 'ECE',
  marks: 47
}
Type 'it' for more
studentProjectDB>
```

### C. MARKS EQUAL TO 85

**Question: How do you find all students whose marks are exactly 85? (\$eq)**

```
db.studentRecords.find({
  marks: { $eq: 85 }
});
```

**SCREENSHOT / OUTPUT**



```
> db.studentRecords.find({ marks: { $eq: 85 } });
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('34374b00aee2791a5d67db37'),
  rollNo: 140,
  name: 'Pallavi',
  age: 21,
  department: 'ME',
  marks: 85
}
studentProjectDB>
```

#### **D. MARKS GREATER THAN OR EQUAL TO 80**

**Question: How do you find all students whose marks are greater than or equal to 80? (\$gte)**

```
db.studentRecords.find({
  marks: { $gte: 80 }
});
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.find({ marks: { $gte: 80 } });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('8f41789677c20cbb63df82a1'),
  rollNo: 108,
  name: 'Hina',
  age: 21,
  department: 'ME',
  marks: 95
}
{
  _id: ObjectId('9d7b687bc34afa032fbead18'),
  rollNo: 110,
  name: 'Jhanvi',
  age: 21,
  department: 'IT',
  marks: 81
}
Type 'it' for more
studentProjectDB>
```

## E. MARKS NOT EQUAL TO 85

Question: How do you find all students whose marks are not equal to 85? (\$ne)

```
db.studentRecords.find({  
  marks: { $ne: 85 }  
});
```

SCREENSHOT / OUTPUT

```
> db.studentRecords.find({ marks: { $ne: 85 } });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('c5ec59fbd922e21fe035b661'),
  rollNo: 106,
  name: 'Falguni',
  age: 21,
  department: 'IT',
  marks: 79
}
Type 'it' for more
studentProjectDB>
```

## F. MARKS GREATER THAN 70 AND AGE LESS THAN 23

Question: How do you find students whose marks are greater than 70 AND age is less than 23? (\$and)

```
db.studentRecords.find({
  $and: [
    { marks: { $gt: 70 } },
    { age: { $lt: 23 } }
  ]
});
```

SCREENSHOT / OUTPUT

```

> db.studentRecords.find({ $and: [{ marks: { $gt: 70 } }, { age: { $lt: 23 } }] });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('c5ec59fbd922e21fe035b661'),
  rollNo: 106,
  name: 'Falguni',
  age: 21,
  department: 'IT',
  marks: 79
}
Type 'it' for more
studentProjectDB>

```

## G. DEPARTMENT IS CSE OR ECE

Question: How do you find students who belong to either CSE OR ECE department? (\$or)

```

db.studentRecords.find({
  $or: [
    { department: "CSE" },
    { department: "ECE" }
  ]
})

```

```
]
});
```

## SCREENSHOT / OUTPUT

```
> db.studentRecords.find({ $or: [{ department: "CSE" }, { department: "ECE" }] });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('9890f5644ee5c3226be7b274'),
  rollNo: 109,
  name: 'Ivaan',
  age: 18,
  department: 'CSE',
  marks: 68,
  status: 'Active'
}
{
  _id: ObjectId('5955602abdcac3807bbc0a88'),
  rollNo: 111,
  name: 'Keshav',
  age: 18,
  department: 'ECE',
  marks: 47
}
Type 'it' for more
studentProjectDB>
```

## H. MARKS NOT GREATER THAN 80

**Question: How do you find students whose marks are NOT greater than 80? (\$not)**

```
db.studentRecords.find({  
  marks: {  
    $not: { $gt: 80 }  
  }  
});
```

**SCREENSHOT / OUTPUT**

```

> db.studentRecords.find({ marks: { $not: { $gt: 80 } } });
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
{
  _id: ObjectId('c5ec59fbd922e21fe035b661'),
  rollNo: 106,
  name: 'Falguni',
  age: 21,
  department: 'IT',
  marks: 79
}
{
  _id: ObjectId('9890f5644ee5c3226be7b274'),
  rollNo: 109,
  name: 'Ivaan',
  age: 18,
  department: 'CSE',
  marks: 68,
  status: 'Active'
}
{
  _id: ObjectId('5955602abdcac3807bbc0a88'),
  rollNo: 111,
  name: 'Keshav',
  age: 18,
  department: 'ECE',
  marks: 47
}
Type 'it' for more
studentProjectDB>

```

## 9. BASIC DISPLAY / VERIFICATION COMMANDS

Show current database

Question: What command is used to show current database?

```
db;
```

#### SCREENSHOT / OUTPUT

```
> db;  
studentProjectDB  
studentProjectDB>
```

### Show collections

Question: What command is used to show collections?

```
show collections;
```

#### SCREENSHOT / OUTPUT

```
> show collections;  
studentRecords  
studentProjectDB>
```

### Show all student records

Question: What command is used to show all student records?

```
db.studentRecords.find();
```

#### SCREENSHOT / OUTPUT

```
> db.studentRecords.find();
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
Type 'it' for more
studentProjectDB>
```

**Show all records in readable format**

**Question: What command is used to show all records in readable format?**

```
db.studentRecords.find().pretty();
```

## SCREENSHOT / OUTPUT

```
> db.studentRecords.find().pretty();
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
Type 'it' for more
studentProjectDB>
```

Count total documents

**Question: What command is used to count total documents?**

```
db.studentRecords.countDocuments();
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.countDocuments();  
44  
studentProjectDB>
```

**Count CSE students**

**Question: What command is used to count cse students?**

```
db.studentRecords.countDocuments({  
  department: "CSE"  
});
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.countDocuments({ department: "CSE" });  
11  
studentProjectDB>
```

**Count ECE students**

**Question: What command is used to count ece students?**

```
db.studentRecords.countDocuments({  
  department: "ECE"  
});
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.countDocuments({ department: "ECE" });
11
studentProjectDB>
```

## Count IT students

Question: What command is used to count it students?

```
db.studentRecords.countDocuments({
  department: "IT"
});
```

### SCREENSHOT / OUTPUT

```
> db.studentRecords.countDocuments({ department: "IT" });
11
studentProjectDB>
```

## Count ME students

Question: What command is used to count me students?

```
db.studentRecords.countDocuments({
  department: "ME"
});
```

### SCREENSHOT / OUTPUT

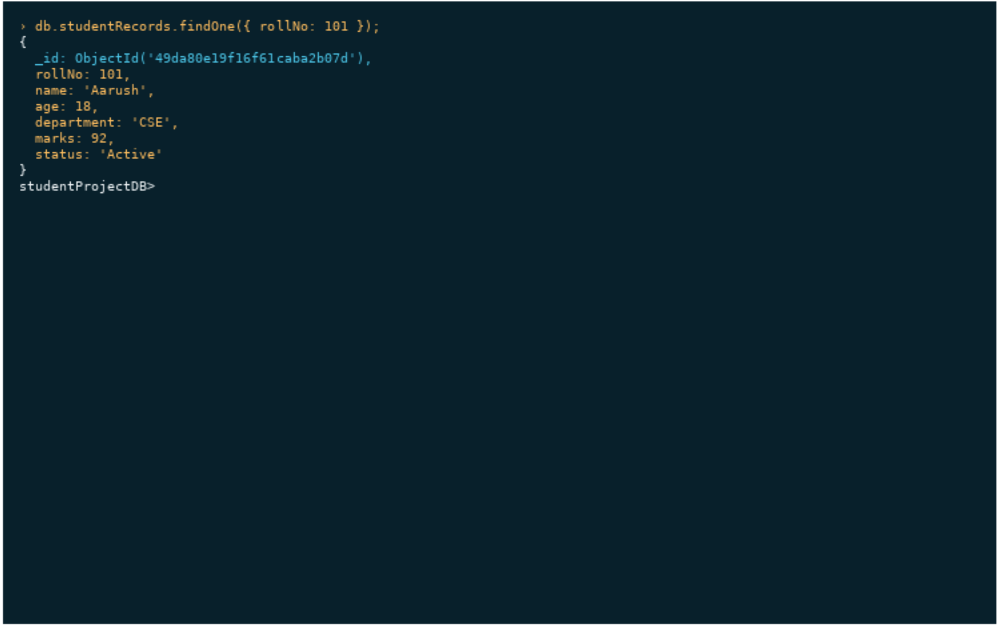
```
> db.studentRecords.countDocuments({ department: "ME" });
11
studentProjectDB>
```

### Find student by Roll No. 101

Question: What command is used to find student by roll no. 101?

```
db.studentRecords.findOne({  
  rollNo: 101  
});
```

#### SCREENSHOT / OUTPUT



```
> db.studentRecords.findOne({ rollNo: 101 });  
{  
  _id: ObjectId('49da80e19f16f61caba2b07d'),  
  rollNo: 101,  
  name: 'Aarush',  
  age: 18,  
  department: 'CSE',  
  marks: 92,  
  status: 'Active'  
}  
studentProject0B>
```

### Find all CSE students

Question: What command is used to find all cse students?

```
db.studentRecords.find({  
  department: "CSE"  
});
```

#### SCREENSHOT / OUTPUT

```

> db.studentRecords.find({ department: "CSE" });
{
  _id: ObjectId('49da80e19f16f61caba2b07d'),
  rollNo: 101,
  name: 'Aarush',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('a2b59de246e7613a59509acc'),
  rollNo: 105,
  name: 'Eshan',
  age: 18,
  department: 'CSE',
  marks: 92,
  status: 'Active'
}
{
  _id: ObjectId('9890f5644ee5c3226be7b274'),
  rollNo: 109,
  name: 'Ivaan',
  age: 18,
  department: 'CSE',
  marks: 68,
  status: 'Active'
}
{
  _id: ObjectId('ba2de915d0ba162e9fe8cfc2'),
  rollNo: 113,
  name: 'Mohit',
  age: 18,
  department: 'CSE',
  marks: 72,
  status: 'Active'
}
{
  _id: ObjectId('fcl2d80efa580ef6ad7f940'),
  rollNo: 117,
  name: 'Qadir',
  age: 18,
  department: 'CSE',
  marks: 58,
  status: 'Active'
}
{
  _id: ObjectId('be82c5a6597c133b3826512b'),
  rollNo: 129,
  name: 'Devika',
  age: 18,
  department: 'CSE',
  marks: 67,
  status: 'Active'
}
Type 'it' for more
studentProjectDB>

```

**Find all ECE students**

**Question: What command is used to find all ece students?**

```
db.studentRecords.find({  
  department: "ECE"  
});
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.find({ department: "ECE" });
{
  _id: ObjectId('6722edffd63466f57afb8d34'),
  rollNo: 103,
  name: 'Chirag',
  age: 18,
  department: 'ECE',
  marks: 85
}
{
  _id: ObjectId('5955602abdcac3807bbc0a88'),
  rollNo: 111,
  name: 'Keshav',
  age: 18,
  department: 'ECE',
  marks: 47
}
{
  _id: ObjectId('5ce305027a3134fb22587ff7'),
  rollNo: 115,
  name: 'Omkar',
  age: 18,
  department: 'ECE',
  marks: 86
}
{
  _id: ObjectId('43c2436ef1697af5a4ec015e'),
  rollNo: 119,
  name: 'Sahil',
  age: 18,
  department: 'ECE',
  marks: 65
}
{
  _id: ObjectId('1ab105bd7793b892140d8592'),
  rollNo: 123,
  name: 'Wamika',
  age: 18,
  department: 'ECE',
  marks: 93
}
{
  _id: ObjectId('d7c6a3d0dd9ef687053a36c9'),
  rollNo: 127,
  name: 'Bhumika',
  age: 18,
  department: 'ECE',
  marks: 52
}
Type 'it' for more
studentProjectDB>
```

## Find all IT students

Question: What command is used to find all it students?

```
db.studentRecords.find({  
  department: "IT"  
});
```

**SCREENSHOT / OUTPUT**

```
> db.studentRecords.find({ department: "IT" });
{
  _id: ObjectId('f8ebaa9930f45b8f0420198f'),
  rollNo: 102,
  name: 'Bhavya',
  age: 21,
  department: 'IT',
  marks: 74
}
{
  _id: ObjectId('c5ec59fbd922e21fe035b661'),
  rollNo: 106,
  name: 'Falguni',
  age: 21,
  department: 'IT',
  marks: 79
}
{
  _id: ObjectId('9d7b687bc34afa032fbeat18'),
  rollNo: 110,
  name: 'Jhanvi',
  age: 21,
  department: 'IT',
  marks: 81
}
{
  _id: ObjectId('df8a188247a735f778cd5ala'),
  rollNo: 118,
  name: 'Ritika',
  age: 21,
  department: 'IT',
  marks: 99
}
{
  _id: ObjectId('d63cdfe85782489fc577855e'),
  rollNo: 122,
  name: 'Varun',
  age: 21,
  department: 'IT',
  marks: 71
}
{
  _id: ObjectId('94a04e74e658a68a56be61cd'),
  rollNo: 126,
  name: 'Advik',
  age: 21,
  department: 'IT',
  marks: 89
}
Type 'it' for more
studentProjectDB>
```

## Find all ME students

Question: What command is used to find all me students?

```
db.studentRecords.find({
  department: "ME"
});
```

### SCREENSHOT / OUTPUT

```
> db.studentRecords.find({ department: "ME" });
{
  _id: ObjectId('fc42f158430cf4fb23e930ce'),
  rollNo: 104,
  name: 'Diya',
  age: 21,
  department: 'ME',
  marks: 63
}
{
  _id: ObjectId('8f41789677c20cbb63df82a1'),
  rollNo: 108,
  name: 'Hina',
  age: 21,
  department: 'ME',
  marks: 95
}
{
  _id: ObjectId('e36bd19c96084e4bd3d33a73'),
  rollNo: 112,
  name: 'Lavanya',
  age: 21,
  department: 'ME',
  marks: 90
}
{
  _id: ObjectId('2715c20794c0e0b8ca47ce83'),
  rollNo: 116,
  name: 'Prisha',
  age: 21,
  department: 'ME',
  marks: 77
}
{
  _id: ObjectId('3c585128379498327137be78'),
  rollNo: 120,
  name: 'Trisha',
  age: 21,
  department: 'ME',
  marks: 83
}
{
  _id: ObjectId('9b90e242ab0b01bfa45e9d10'),
  rollNo: 124,
  name: 'Yash',
  age: 21,
  department: 'ME',
  marks: 76
}
Type 'it' for more
studentProjectDB>
```

**Thank You**